

Modelling and prediction of extreme rainfall events

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Overview

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- Experiments
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- Appendix

Introduction

Problem Statement

The objective of this work is to develop and systematically evaluate deep learning-based approaches for extreme rainfall prediction using historical multi-station time-series data.

The focus is on:

- Modeling temporal dependencies in rainfall dynamics
- Handling heteroscedastic and heavy-tailed behavior
- Improving predictive performance for high-intensity rainfall events

Data Overview

Data Available

- Daily rainfall measurements (in mm) for 293 stations of West Bengal from year 1901 to 2020
- Latitude and longitude of the station

Why this is a Hard Problem

- Rainfall is intermittent (many low/zero days)
- Extreme events are rare but high-impact
- Variance changes over time (heteroscedasticity)
- Spatial heterogeneity across stations
- Only past rainfall data is present

Literature Survey

Paper Reference	Architecture	Short Description	Data Used	Results
Case Study 1	Neural Network Autoregression (NNAR)	A multilayer feed-forward neural network that uses both non-seasonal and seasonal lags to capture monsoon oscillations.	IMD records (1901–2018) for Birbhum and North 24 Parganas districts, West Bengal.	Established predictive power over traditional ARIMA models; however, out-of-sample RMSE remained high during extreme monsoon years.
Case Study 2	Adaptive Sliding Window (ASWA)	A dynamic segmentation algorithm that uses expert-defined weather categories (change points) to partition data into variable-sized windows.	Daily rainfall and river flow data from Malaysia (categorized into classes: No rain, Light, Moderate, and Heavy).	Achieved optimal segmentation of noisy signals, improved error detection, and reduced dimensionality while preserving key features.
Case Study 3	DWT-LSTM	Combines Discrete Wavelet Transform (DWT) to denoise the signal into "base" and "noise" components, with an LSTM modeling the noise residuals.	Daily rainfall time series from Philippines subcatchments.	Effectively captured stochastic nature with NSE values between 0.94 and 0.98 and MAPE values approaching zero.
Case Study 4	Vanilla LSTM / GRU / BiLSTM	A comparative study of various LSTM variants, including stacked, CNN-hybridized, and attention-based models.	30-year monthly rainfall record (1994–2023) from Penang Island, Malaysia.	Traditional RNN and GRU models surprisingly outperformed more complex hybrid architectures for monthly univariate data.

Literature Survey

Case Study 5	GRU-LSTM and CNN-GRU	A comparative study of various LSTM variants, including stacked, CNN-hybridized, and attention-based models.	Monthly precipitation records (2000–2023) for Istanbul and Konya, Turkey.	The GRU-LSTM model generally outperformed CNN-GRU, suggesting recurrent gates are better than convolutions for purely univariate series.
Case Study 6	SARITA (DConvLSTM with Attention)	Integrates an attention-guided spatial correlation mechanism using Local Moran’s I within a Deformable ConvLSTM.	Hourly rainfall from IMD stations in Jammu, Srinagar, and Ladakh (2019–2024).	Achieved a 25–35% improvement in hourly forecasts over standard ConvLSTM and improved anomaly detection by 3.5%.
Case Study 7	EWT-LSTM	Uses Empirical Wavelet Transform (EWT) to decompose Indian Summer Monsoon Rainfall (ISMR) into multiple modes processed by an LSTM.	ISMR rainfall sums for the June–September period from 1871 to 2022.	Achieved a performance parameter (PP) of 0.86, notably higher than previous methods like Generalized Regression Neural Networks.

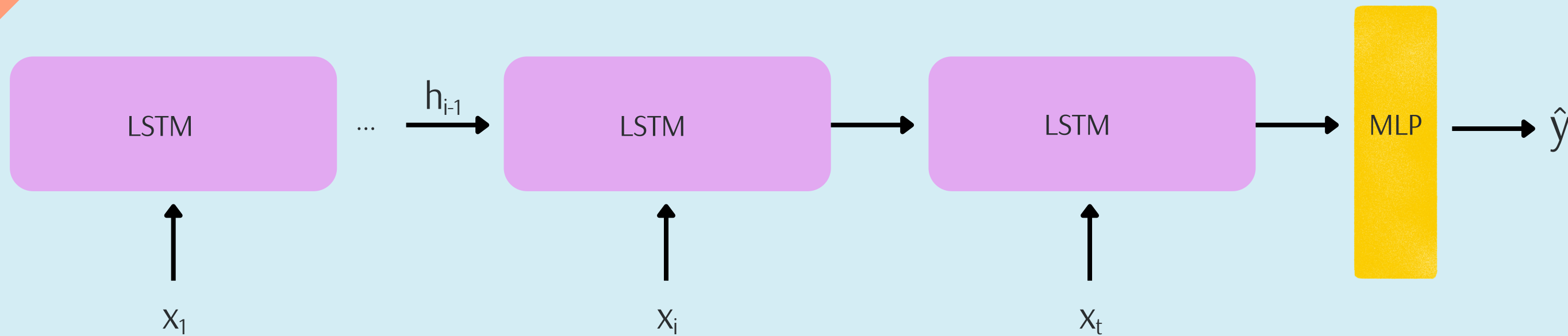
Reference	Full Paper Title
Case Study 1	<i>Statistical and Machine Learning-based Modeling of Rainfall Variability in Birbhum and North 24 Parganas, West Bengal (1901–2018).</i>
Case Study 2	<i>Yahmed, Y. B., Bakar, A. A., Hamdan, A. R., Ahmed, A., & Abdullah, S. M. S. (2015). "Adaptive Sliding Window Algorithm for Weather Data Segmentation." Journal of Theoretical and Applied Information Technology, 80(2), 322-332.</i>
Case Study 3	<i>Denoising and Forecasting Daily Rainfall Using Discrete Wavelet Transform and Long Short-Term Memory Networks.</i>
Case Study 4	<i>Comparative Analysis of RNN, LSTM, and GRU Architectures for Monthly Rainfall Prediction in Penang Island, Malaysia.</i>
Case Study 5	<i>Univariate Precipitation Forecasting in Semi-Arid and Maritime Climates: A Comparison of CNN-GRU and GRU-LSTM Models.</i>
Case Study 6	<i>SARITA: Attention-Guided Deformable ConvLSTM for Hourly Rainfall Forecasting in the Himalayan Regions of Jammu & Kashmir.</i>
Case Study 7	<i>Empirical Wavelet Transform-based Deep Learning for Indian Summer Monsoon Rainfall (ISMR) Prediction.</i>

Experiments

Experimental Setup:

- Input: Multivariate time series per station, consisting of rainfall observations and optional meta data.
- Output: One-step or multi-step ahead rainfall prediction.
- Training : Supervised learning with context windows.
- Evaluation focus: Overall rmse,mae and performance on rainy vs non-rainy days, and tail behavior for extreme rainfall events.

Experiment1: Baseline Lstm



We model rainfall as a sequence forecasting problem:

$$y_{(t+h)} = f(x(t-w : t))$$

Using a standard LSTM:

$$h_{(t)} = \text{LSTM}(x_{(t)}, h_{(t-1)})$$

$$\hat{y}_{(t+h)} = W \cdot h_{(t)} + b$$

where,

$$x(t) = [R(t) | \sin(\text{DOY}(t)) | \cos(\text{DOY}(t)) | \sin(\text{Month}(t)) | \cos(\text{Month}(t)) | \sin(\text{RelYear}(t)) | \cos(\text{RelYear}(t))]$$

Loss Function

We use Mean Squared Error (MSE):

$$\text{MSE} = (1 / N) \times \sum (y - \hat{y})^2$$

Results:

Model	Category	RMSE	MAE	Bias
Baseline LSTM	Monsoon	23.079325	13.090454	-0.419241
	Non-Monsoon	8.711857	2.716992	-0.059962
	Overall	15.846863	6.696569	-0.197792

Observation:

- Model underpredicts large rainfall values
- Negative bias observed
- High errors during monsoon season
- This suggests insufficient emphasis on extreme events during training.

Experiment2: Weighted Loss

Hypothesis

- If larger prediction errors are assigned higher importance during training, the model will better learn extreme rainfall patterns.

The weighted loss can be written as:

$$L_{\text{FocalMSE}} = (1 - \exp^{-|e|})^\gamma \cdot e^2,$$

where,

- $e = \hat{y} - y$
- $\gamma > 0$ controls focusing strength

Interpretation

- When the error is small, $(1 - \exp^{-|e|})^\gamma$, becomes close to zero.
- When the error is large, the weight moves toward one.
- Large errors receive stronger emphasis

Experiments 3: Seasonal weighted loss

The weighted loss with seasonal weighting:

$$L_{\text{FocalMSE}} = \text{seasonal_weight} * (1 - \exp^{-|e|})^\gamma * e^2$$

where,

- $e = \hat{y} - y$
- $\gamma > 0$ controls focusing strength
- Seasonal weight :
 - monsoon seasons = 3
 - otherwise = 1

Results:

Model	Category	RMSE	MAE	Bias
LSTM (Baseline)	Monsoon	23.079325	13.090454	-0.419241
	Non-Monsoon	8.711857	2.716992	-0.059962
	Overall	15.846863	6.696569	-0.197792
FocalMSE LSTM	Monsoon	23.0347	13.2425	0.0553
	Non-Monsoon	8.7213	2.9568	0.1989
	Overall	15.8251	6.9027	0.1438
Seasonal Focal Loss LSTM	Monsoon	21.0308	12.03476	0.3756
	Non-Monsoon	8.5207	2.8859	0.1633
	Overall	13.8227	6.5993	0.2447

Experiment 4: Multi-Scale LSTM with Attention

Hypothesis

Extreme rainfall events (storms) exhibit:

- Build-up phase (short-term intensification)
- Sustained phase
- Decay phase
- Influence of longer climatic context

A single fixed context window is insufficient to capture both short-term storm dynamics and long-term seasonal structure.

Therefore, we introduce multiple temporal scales.

○



Experiment 4: Multi-Scale LSTM with Attention

Table 3: Adapted sliding window algorithm (ASWA)

Input : Processed weather-related time series sequences $T=t_1, t_2, \dots, t_n$

Output: Number of segmented weather cases (Sequences).

```
1  Begin
2  Initialize  $start\_point = t_i$ ,  $point = start\_point$ ,
    $next = start\_point + 1$  ; and  $end\_point = next$ ;
3  for each  $t_i$  in  $T$ ,  $i = 1..n$ 
4      Set  $error = 0$ 
5      Read data at time  $t_i$ 
6      if ( $point$  is not equal to  $next$  and  $error$  equal to 0) // weather level change (change point)
7          Create a segment start from  $start\_point$  to  $end\_point$  // add into window slices set
8          Set  $point = next + 1$ ; and  $next = point + 1$ 
9          Set the  $start\_point = point$ ;  $end\_point = next$ ; and  $error = 0$ 
10     else
11         Set  $next = i$ ;
12         Set  $end\_point = next + 1$ ;
13         Set  $error = 1$ 
14     end if
15 end for
16 End
```

ASWA is not:

- Learning the optimal context window
- Optimizing window length for prediction

Experiment 4: Multi-Scale LSTM with Attention

Multi-Scale Context Windows

We use parallel LSTMs over different window lengths:

- Scales={3,7,14,30}
- 3-day → immediate storm build-up
- 7-day → short-term accumulation
- 14-day → medium-term persistence
- 30-day → seasonal context

Each scale produces a hidden representation:

- $H^{(s)} \in \mathbb{R}^{T \times d}$

Experiment 4: Multi-Scale LSTM with Attention

Architecture:

I. Temporal Self-Attention (Within Scale)

For each scale s :

$$\alpha_t^{(s)} = \text{softmax}(\mathbf{W}_s \mathbf{H}_t^{(s)})$$

$$\mathbf{z}^{(s)} = \sum_t \alpha_t^{(s)} \mathbf{H}_t^{(s)}$$

This allows the model to:

- Focus on important timesteps
- Emphasize peak rainfall moments

Experiment 4: Multi-Scale LSTM with Attention

Architecture:

2. Cross-Scale Attention (Across Scales)

Representations across scales:

$$\mathbf{Z} = [\mathbf{z}^{(3)}, \mathbf{z}^{(7)}, \mathbf{z}^{(14)}, \mathbf{z}^{(30)}]$$

Cross-scale attention:

$$\beta_s = \text{softmax}(\mathbf{W}_c \mathbf{z}^{(s)})$$

$$\mathbf{z}_{\text{final}} = \sum_s \beta_s \mathbf{z}^{(s)}$$

This learns which temporal scale is most relevant for a given prediction.

Results:

Model	Category	RMSE	MAE	Bias
LSTM (Baseline)	Monsoon	23.079325	13.090454	-0.419241
	Non-Monsoon	8.711857	2.716992	-0.059962
	Overall	15.846863	6.696569	-0.197792
Seasonal Focal Loss	Monsoon	21.0308	12.03476	0.3756
	Non-Monsoon	8.5207	2.8859	0.1633
	Overall	13.8227	6.5993	0.2447
Multiscale LSTM with Attention	Monsoon	17.0612	9.3279	0.2348
	Non-Monsoon	6.7386	1.6928	0.2567
	Overall	10.5587	4.5802	0.2483

Experiment 5: ConvLSTM for Spatio-Temporal Rainfall Forecasting

Hypothesis

Rainfall at a given grid location and time is significantly influenced by spatio-temporal rainfall patterns in neighboring regions over preceding days.

Objective:

Forecast next-day gridded rainfall using historical station data.



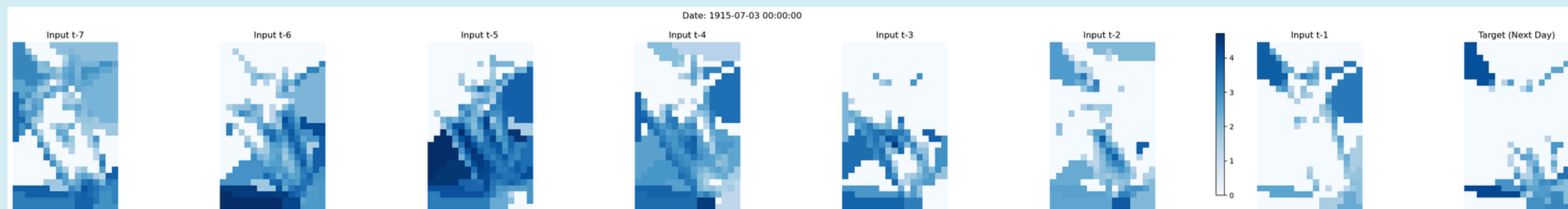
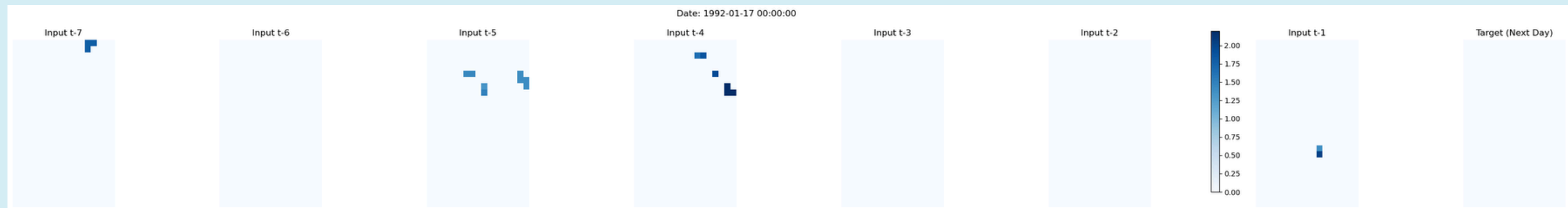
Data processing

Spatial Gridding

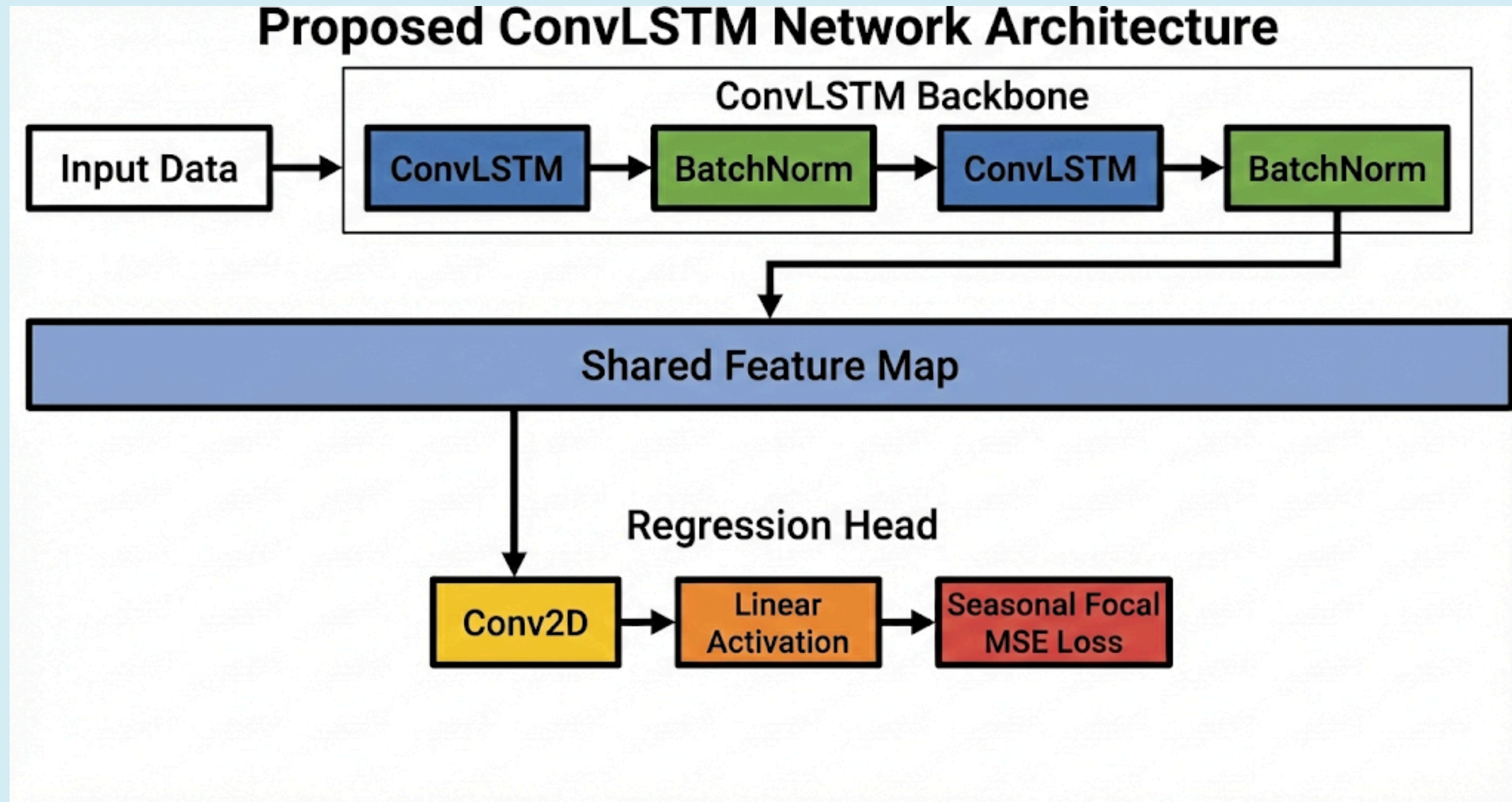
- Interpolation: Linear + Nearest fill
- Grid resolution: 0.25°
- Output shape: (Time, Height, Width)

Thresholding

- IMD rain definition: ≥ 2.5 mm
- Values below threshold set to 0



Model architecture



Results:

Model	Category	RMSE	MAE	Bias
ConvLSTM	Monsoon	20.6467	11.821899	-0.864164
	Non-Monsoon	7.538165	2.033314	-0.010447
	Overall	12.1383	5.46124	-0.31564
Seasonal Focal Loss	Monsoon	21.0308	12.03476	0.3756
	Non-Monsoon	8.5207	2.8859	0.1633
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Experiment 6: Regression via classification

Hypothesis:

Modeling rainfall as a sequence of discretized intensity tokens and training a causal transformer with cross-entropy will better capture temporal dependencies and seasonal structure compared to traditional LSTM regression.

Key Idea:

Reformulate regression as:

$$p(z_{t+1} \mid z_{1:t}, \text{DOY})$$

Where:

- z_t = rainfall bin token
- DOY = seasonal embedding

Data preparation:

Rain > 0

```
count      3.959148e+06
mean       5.485089e+00
std        1.714293e+01
min         0.000000e+00
50%         0.000000e+00
75%         1.400000e+00
75.5%       1.800000e+00
76%         2.000000e+00
76.5%       2.200000e+00
77%         2.500000e+00
77.5%       2.600000e+00
78%         3.000000e+00
78.5%       3.300000e+00
79%         3.600000e+00
79.5%       4.000000e+00
80%         4.300000e+00
80.5%       4.700000e+00
81%         5.100000e+00
81.5%       5.300000e+00
82%         5.800000e+00
82.5%       6.400000e+00
83%         6.800000e+00
83.5%       7.300000e+00
84%         7.600000e+00
84.5%       8.200000e+00
...
99%         8.260000e+01
99.5%       1.092000e+02
99.95%      2.169427e+02
max         8.845000e+02
Name: rainfall, dtype: float64
```

```
count      1.121149e+06
mean       1.936966e+01
std        2.772812e+01
min         1.000000e-01
10%         1.300000e+00
15%         2.000000e+00
20%         2.500000e+00
25%         3.500000e+00
30%         4.400000e+00
35%         5.500000e+00
40%         6.800000e+00
45%         8.100000e+00
50%         1.000000e+01
75%         2.410000e+01
75.5%       2.450000e+01
76%         2.500000e+01
76.5%       2.540000e+01
77%         2.570000e+01
77.5%       2.620000e+01
78%         2.680000e+01
78.5%       2.740000e+01
79%         2.790000e+01
79.5%       2.840000e+01
80%         2.920000e+01
80.5%       3.000000e+01
81%         3.050000e+01
81.5%       3.100000e+01
82%         3.180000e+01
82.5%       3.250000e+01
83%         3.300000e+01
```

```
86.5%      3.940000e+01
87%         4.040000e+01
87.5%       4.140000e+01
88%         4.270000e+01
88.5%       4.390000e+01
89%         4.520000e+01
89.5%       4.650000e+01
90%         4.820000e+01
90.5%       4.980000e+01
91%         5.100000e+01
91.5%       5.300000e+01
92%         5.460000e+01
92.5%       5.680000e+01
93%         5.900000e+01
93.5%       6.140000e+01
94%         6.380000e+01
94.5%       6.680000e+01
95%         7.020000e+01
95.5%       7.420000e+01
96%         7.820000e+01
96.5%       8.310000e+01
97%         8.890000e+01
97.5%       9.580000e+01
98%         1.041000e+02
98.5%       1.156000e+02
99%         1.334000e+02
99.5%       1.654000e+02
99.95%      2.860000e+02
max         8.845000e+02
```

Data preparation:

Discretization Strategy

Instead of uniform binning:

- 1 bin $\rightarrow$ rainfall = 0
- 5% percentile bins up to 75%
- 1% bins from 76–94%
- 0.5% bins from 94.5–99.5%
- 1 overflow bin

Total: 48 bins

Rationale:

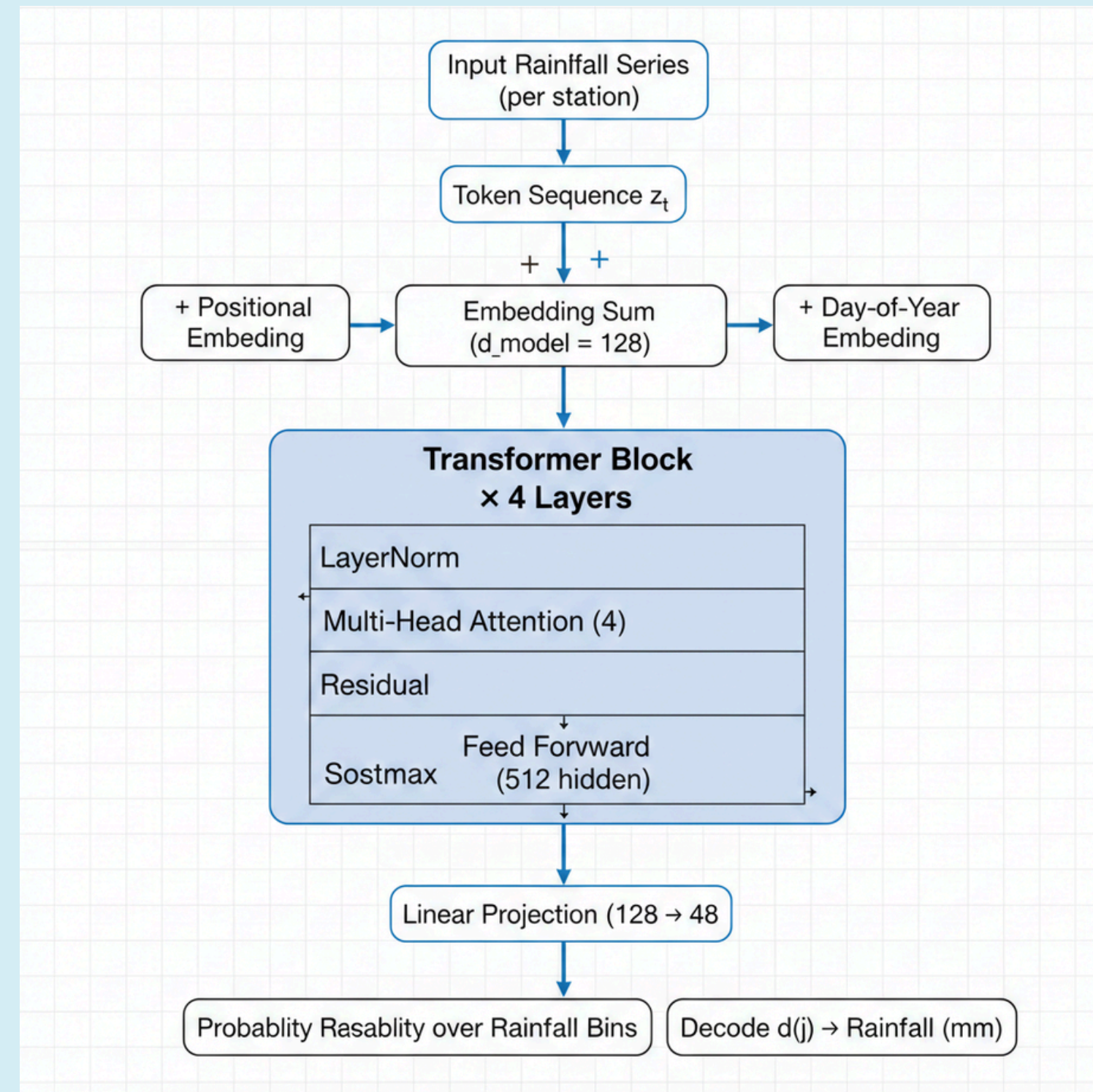
- Higher resolution in dense regions
- Controlled resolution in extreme tail
- Avoid sparse class explosion

$$q(x) = \begin{cases} 0 & \text{when } x = 0 \\ j & \text{when } b_{(j-1)} < x \leq b_{(j)} \\ B & \text{when } x > b_{(B-1)} \end{cases}$$

$$d(j) = (b_{(j-1)} + b_{(j)}) / 2$$



Architecture



Results:

Model	Category	RMSE	MAE	Bias
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	Overall	10.5587	4.5802	0.2483
Transformer Architecture	Monsoon	28.1546	12.9591	-10.1502
	Non-Monsoon	9.229	1.9294	-1.786
	Overall	18.5321	5.9992	-4.9424

Future Work

Hypothesis

H_I: Attention-based spatial aggregation will improve representation of dynamic inter-station dependencies.

Future Work

The Problem: Purely autoregressive models (using only past rain) often collapse to "mean drizzle" because they treat rainfall as a closed system.

The Solution: A three-component DL pipeline:

1. Spatio-Temporal Encoder (GNN) : Lifts station data and atmospheric drivers into a high-dimensional latent state.
2. Koopman Linear Layer : Evolves the latent state using a learned transition matrix K and a control matrix B

$$y_{t+1} = Ky_t + Bu_t$$

3. Decoder : Maps the state to some distribution, effectively handling zero-inflation (dry days) and heavy tails (extreme storms)



Thank You